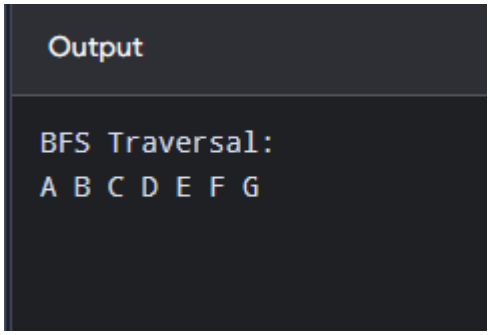
Lab experiment

Breadth First Search (BFS)

Coding:

```
# Breadth First Search (BFS)

from collections import deque

# Graph representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

# BFS function
def bfs(start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbor in graph[node]:
                if neighbor not in visited:
                    queue.append(neighbor)

# Main program
print("BFS Traversal:")
bfs('A')
```

Output: